Data Engineering Assignment 1: Data Pipeline Design

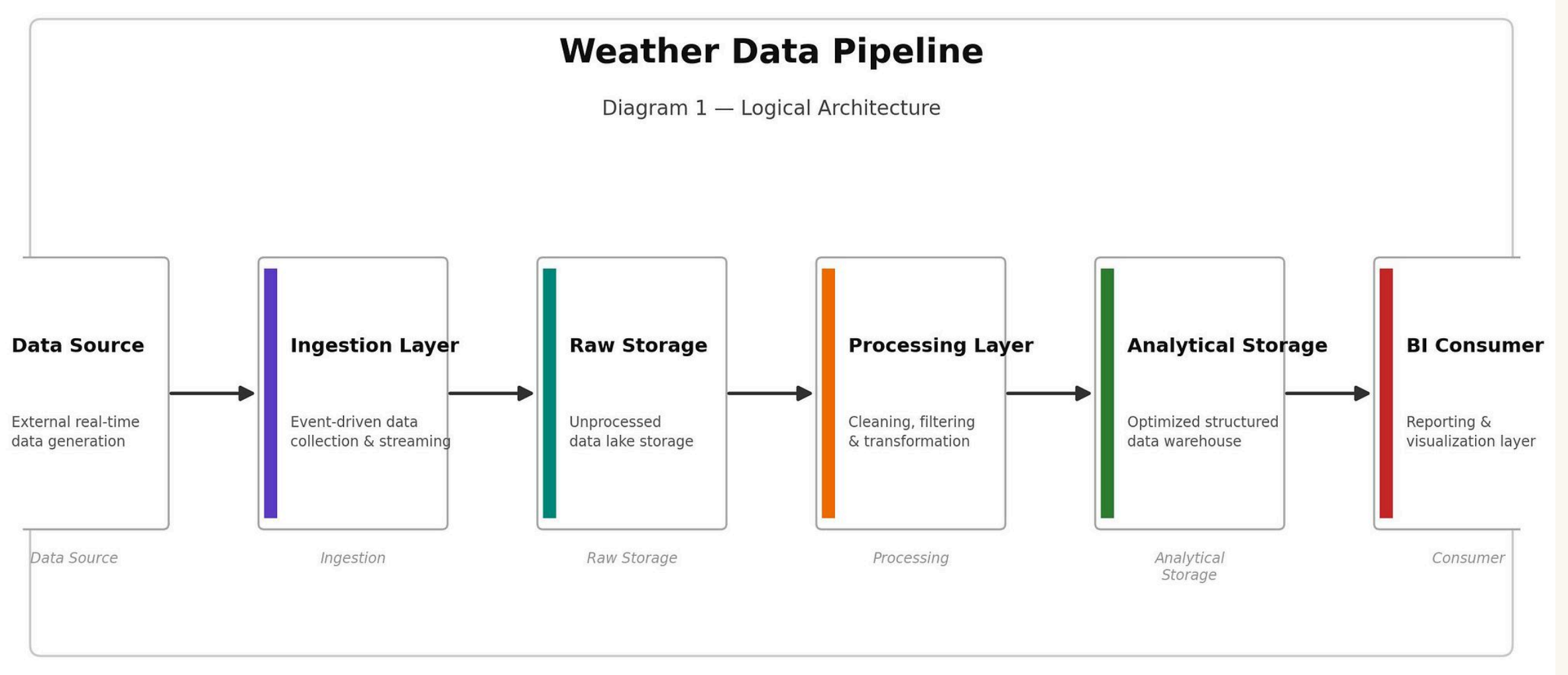
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Selected Source

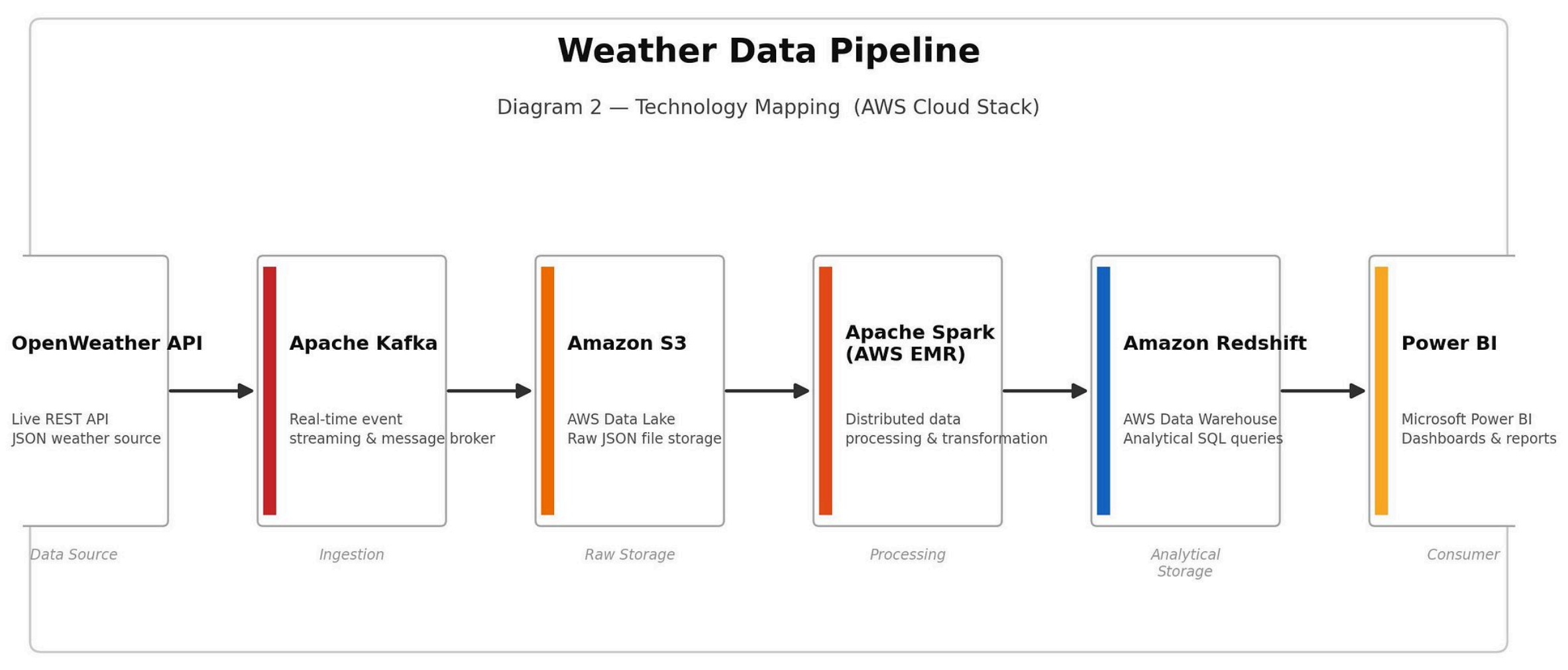
01 **System/Source:** OpenWeather API (Current Weather Data API)

02 **Type of data produced:** JSON format containing nested weather attributes such as temperature, humidity, wind speed, atmospheric pressure, geographical coordinates, and Unix timestamps.

Logical Architecture



Technology Mapping



Tech Stack Used:

01 OpenWeather API

02 Apache Kafka

03 Amazon S3

04 Apache Spark (AWS EMR)

05 Amazon Redshift

06 Power BI

Component Explanation

OpenWeather API: Acts as the real-time source of raw weather data in JSON format.

Apache Kafka: Used to stream the incoming API events reliably into the pipeline as a message queue, ensuring no data is lost even during high traffic.

Amazon S3: Serves as the AWS data lake to store all raw, unprocessed JSON files before any transformation—acts as a safety net for future reprocessing.

Apache Spark (AWS EMR): Used on AWS Elastic MapReduce for distributed processing—cleans, filters, flattens nested JSON, and converts data into a structured tabular format at scale.

Amazon Redshift: An AWS columnar data warehouse used to store the transformed, structured data so it can be queried efficiently using SQL for analytics.

Power BI: Used by business analysts to connect to Amazon Redshift and build interactive dashboards and visual reports that track weather trends over time.

Question Log

01 If the OpenWeather API gives us live data, why can't Power BI just connect directly to the API instead of going through Kafka and Redshift?

02 Why do we need Apache Kafka at all? Can't we just use a Python script to call the API directly every hour?

03 Why is Amazon S3 needed as raw storage if Apache Spark is going to process the data anyway — why not skip S3 and process directly?

04 What happens if the OpenWeather API changes its JSON field names—will our entire Spark processing job on EMR break silently?

05 What is the difference between Amazon S3 and Amazon Redshift—aren't both just storage?

Short Architecture Explanation

Why does data move from one component to the next?

Data originates at the OpenWeather API, which publishes live weather readings in JSON format. It flows into Apache Kafka because Kafka acts as a reliable message broker — it decouples the data source from the rest of the pipeline so that even if a downstream system is slow, no data is dropped. From Kafka, the raw JSON events are landed into Amazon S3 to create a permanent archive of all original data on AWS; this ensures we can always reprocess if something goes wrong later.

Next, Apache Spark running on AWS EMR reads from S3 and performs the heavy lifting — it cleans missing fields, flattens deeply nested JSON into flat columns, and converts timestamps into readable formats. Spark on EMR is used here because it can process large volumes of data in a distributed and cost-effective manner on AWS infrastructure. The clean, flat data is then written into Amazon Redshift, a columnar data warehouse on AWS optimized for fast analytical SQL queries at scale. Finally, Power BI connects directly to Amazon Redshift to let business users build interactive charts, trend graphs, and weather reports — without ever needing to write code themselves.